IMO 2026 — Problem 5

Solution by Claude Fable 5

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Source: `results/imo-2026/claude-fable-5/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Approaches tried

- **two-point-pinch** — worked (round 1, APPROVED). Core identity f(f(y)) = 2f(y) − y from x = f(y); orbit AP gives f   id; second substitution x = f(y) into the right inequality gives the pinch |d(a) − d(b)| (a—b)²/(4 · min(a,b)); telescoping subdivision forces d constant. Every algebraic identity re-verified symbolically by the reviewer.
- **marching-orbits** — worked (round 1, APPROVED as an independent second proof). Same core identity and AP orbits; mismatched positive d-values p < q refuted by marching the p-orbit to within p of a far q-orbit point where the right inequality's slack Δ = 4y_n(p—q) + (S+p)² − 4Sq < 0; mixed {0,c} case killed by fixed-point propagation (Lemma 5 interval, quadratic root computation verified) and an exclusion zone |s—a| 2√(ac) that the c-orbit (step c < interval length 4√(a c)) must enter. All constants and inductions checked.
- **tangent-envelope** — not built in round 1 (reserve; open structural gap on density of range(f) anchors). Superseded by the solve.

Current best

Problem solved. Answer: **exactly the functions f(x) = x + c, c   0 a constant.** Two independent complete proofs exist (`approaches/two-point-pinch.md`, `approaches/marching-orbits.md`). Certified shared lemmas: `lemmas/core-identity-ap-orbits.md`, `lemmas/two-point-pinch-bound.md`.

Full proof

Problem. Determine all functions f: _{>0} → _{>0} such that

√((x^2 + f(y)^2)/2)   (f(x) + y)/2   √(x · f(y)) for all x, y > 0. ( )

Answer. Exactly the functions f(x) = x + c, where c   0 is a constant.

Throughout, we use the fact that for real numbers A, B   0, the inequality A   B is equivalent to A^2 B^2 (squaring is strictly increasing on [0, ∞); cf. knowledge base entry "Standard inequalities (QM-AM-GM equality cases)"). We invoke this each time we square, after checking both sides are nonnegative.

Part I. Sufficiency: every f(x) = x + c with c   0 satisfies ( )

First, f is well defined as a map _{>0} → _{>0}: for x > 0 and c   0 we have f(x) = x + c > 0. (If c < 0, then for 0 < x   −c we would have f(x)   0, violating the codomain; so no f(x) = x + c with c < 0 is even a candidate — a domain/codomain fact, independent of ( ).)

Fix x, y > 0 and write f(y) = y + c, f(x) = x + c.

Left inequality. Both sides of √((x^2 + (y+c)^2)/2)   (x + c + y)/2 are positive, so the inequality is equivalent to its square: 2(x^2 + (y+c)^2)   (x + y + c)^2. Expanding,

2(x^2 + (y+c)^2) − (x + y + c)^2 = x^2 + y^2 + c^2 − 2xy − 2xc + 2yc = (x − (y + c))^2 0,

where the last identity is checked by expanding (x − y − c)^2. Hence the left inequality holds.

Right inequality. Both sides of (x + y + c)/2   √(x(y+c)) are nonnegative, so it is equivalent to its square: (x + y + c)^2 4x(y + c). Writing t = y + c > 0,

(x + t)^2 − 4xt = (x − t)^2 = (x − (y + c))^2 0.

So every f(x) = x + c with c   0 satisfies ( ), with equality throughout exactly when x = y + c. (Part I)

Part II. Uniqueness: any f satisfying ( ) equals x   x + c for some constant c   0

Let f: _{>0} → _{>0} satisfy ( ). Define d: _{>0} →   by d(y) := f(y) − y.

Step 1 (Core identity): f(f(y)) = 2f(y) − y for all y > 0.

Fix y > 0. Since f(y) > 0, we may substitute x = f(y) into ( ). Then the leftmost term is √((f(y)^2 + f(y)^2)/2) = √(f(y)^2) = f(y) (as f(y) > 0), and the rightmost term is √(f(y) · f(y)) = f(y). So ( ) reads

f(y)   (f(f(y)) + y)/2   f(y)

(f(f(y)) is defined because f(y) > 0). A chain of inequalities with equal endpoints forces equality throughout:

(f(f(y)) + y)/2 = f(y), i.e. **f(f(y)) = 2f(y) − y**. (1)

Step 2 (f   id): d(y)   0 for all y > 0.

Fix y > 0 and define the forward orbit y := y, y_{n+1} := f(y_n) for n   0.

Claim A: y_n > 0 for every n   0. Induction on n. Base: y = y > 0. Step: if y_n > 0, then y_n is in the domain of f, and y_{n+1} = f(y_n) > 0 by the codomain.

Claim B: y_{n+2} = 2y_{n+1} − y_n for every n   0. By Claim A, y_n > 0, so identity (1) applies at the point y_n: y_{n+2} = f(f(y_n)) = 2f(y_n) − y_n = 2y_{n+1} − y_n.

Claim C: y_n = y + n · d(y) for every n   0. Two-step induction (KB "Linear recurrences": characteristic polynomial (− 1)², arithmetic progressions): for n = 0, y = y; for n = 1, y = f(y) = y + d(y). Assuming the formula for n and n + 1,

y_{n+2} = 2y_{n+1} − y_n = 2(y + (n+1)d(y)) − (y + n · d(y)) = y + (n+2)d(y).

Now suppose, for contradiction, that d(y) < 0. By the Archimedean property of , choose an integer n > y / (−d(y)). Then

y_n = y + n · d(y) < y + (y / (−d(y))) · d(y) = 0,

contradicting Claim A. Hence **d(y)   0, i.e. f(y)   y, for all y > 0**. (2)

Step 3 (Two-point pinch): |d(a) − d(b)| (a − b)² / (4 · min(a, b)) for all a, b > 0.

Fix y, y > 0. Since f(y) > 0, substitute x = f(y) into the right inequality of ( ):

(f(f(y)) + y)/2   √(f(y) · f(y)).

Both sides are nonnegative (the left side is positive because f(f(y)) > 0 and y > 0), so squaring gives the equivalent

(f(f(y)) + y)²   4 f(y) f(y). (3)

By (1), f(f(y)) = 2f(y) − y = y + 2d(y). Substituting into (3) and writing u := d(y), v := d(y):

(y + y + 2u)²   4(y + u)(y + v). (4)

Expanding both sides,

(y + y + 2u)² = (y + y)² + 4uy + 4uy + 4u², 4(y + u)(y + v) = 4y y + 4y v + 4uy + 4uv,

so, using (y + y)² − 4yy = (y − y)²,

LHS − RHS = (y − y)² + 4uy + 4u² − 4y v − 4uv = (y − y)² + 4(y + u)(u − v).

Since y + u = f(y), inequality (4) says exactly

(y − y)² + 4 f(y) · (d(y) − d(y))   0. (5)

Because f(y) > 0, dividing by 4f(y) and rearranging gives d(y) − d(y)   (y − y)² / (4 f(y)). By (2), f(y)   y > 0, so

d(y) − d(y)   (y − y)² / (4 y). (6)

Inequality (6) holds for ALL pairs y, y > 0. Exchanging roles,

d(y) − d(y)   (y − y)² / (4 y). (7)

For arbitrary a, b > 0, applying (6) and (7) with {y, y } = {a, b} in the two orders bounds both d(a) − d(b) and d(b) − d(a) above by (a − b)²/(4 · min(a, b)) — in each case the denominator is 4 times one of a, b, and min(a,b)   each — hence

|d(a) − d(b)| (a − b)² / (4 · min(a, b)) for all a, b > 0. (8)

(No assumption on the sign or size of d(y) is used beyond f(y)   y > 0, which always holds by (2).)

Step 4 (Subdivision kills the quadratic error): d is constant.

Fix 0 < a < b and let k   1 be an integer. Set t_i := a + i(b − a)/k for i = 0, 1, …, k, so t = a, t_k = b, each t_i   a > 0, and t_{i+1} − t_i = (b − a)/k. By the triangle inequality (telescoping d(a) − d(b) = Σ_{i=0}^{k−1} (d(t_i) − d(t_{i+1}))) and (8) applied to each consecutive pair:

|d(a) − d(b)| Σ_{i=0}^{k−1} (t_{i+1} − t_i)² / (4 · min(t_i, t_{i+1})).

For each i, min(t_i, t_{i+1}) = t_i   a, so each of the k summands is at most ((b − a)/k)² / (4a), hence

|d(a) − d(b)| k · ((b − a)/k)² / (4a) = (b − a)² / (4ak). (9)

(9) holds for every positive integer k, while its left side does not depend on k. If |d(a) − d(b)| =   > 0, choose (Archimedean property) an integer k > (b − a)²/(4a); then (9) gives   (b − a)²/(4ak) < , a contradiction. Hence **d(a) = d(b)**.

Since 0 < a < b were arbitrary (and a = b is trivial), d is constant: d   c for some real c. By (2), c = d(1)   0. Therefore

f(x) = x + c for all x > 0, with c   0 a constant. (Part II)

Conclusion

Part II shows every solution of ( ) is of the form f(x) = x + c with c   0; Part I verifies that every such function is a solution (both inequalities reduce to the perfect square (x − y − c)²   0), and functions x + c with c < 0 fail the codomain requirement. Hence the set of all solutions is exactly

{ f(x) = x + c : c   0 constant }.

Reviewer note (round 1): every load-bearing identity above — the two sufficiency slacks, the pinch identity (5), and Step 1's collapse — was re-verified symbolically (sympy). An independent second complete proof of the same characterization, by asymptotic orbit-marching plus fixed-point exclusion zones, is in `approaches/marching-orbits.md` and was also verified in full.